

SolarMax — Wiring Reference

All pin assignments are defined in `include/config.h`. The tables below are the authoritative per-pin detail; use them with the colored diagrams in `wiring_diagram.pdf` and `wiring_breadboard.pdf`.

Related files

- `breadboard_placement.md` — column-by-column placement map (single source of truth for the breadboard).
- `wiring_breadboard.pdf` — **build from this**. Full breadboard layout, every wire drawn.
- `wiring_diagram.pdf` — schematic overview (power distribution, signal map, voltage-divider details). Good for understanding *what connects to what*.

Rebuild any diagram with `pdflatex <file>.tex`.

Datasheet verification status

Checked against the PDFs in `../datasheets/` on 2026-06-19.

Part	Datasheet	Result
ESP32 pin map (input-only 34–39, ADC1, no strapping pins used)	<code>esp32_datasheet_en.pdf</code>	✓ confirmed
BTS7960 pinout + 3.3 V logic + 6–27 V motor supply	<code>BTS7960 Motor Driver.pdf</code>	✓ confirmed (resolved the level-shifter question)
DS3231 I2C address 0x68	<code>DS3231.pdf</code>	✓ confirmed
Potentiometer (P160 rotary, 3-terminal)	<code>p160.pdf</code>	✓ confirmed
Anemometer = QS-FS powered sensor, not a reed switch	<code>C2192-002-datasheet.pdf</code>	! wiring corrected — see anemometer section
LDR / photocell	<code>SEN-09088.pdf</code>	! image-only PDF, not machine-readable; wiring is standard 2-terminal photocell
Solar panel (Renogy 50 W)	<code>50D-SS-Datasheet.pdf</code>	i this is the tracked panel, not control wiring

Motor: 5882-50ZY identified from the model number (stall $\sim 15\text{--}19\text{ A}$, max $200\text{ kg}\cdot\text{cm}$). **Confirm the gear-ratio/RPM variant ordered** — only the high-ratio (low-RPM) versions meet the $20\text{ N}\cdot\text{m}$ wind-holding requirement; the fast 440 RPM version is $\sim 10\times$ too weak.

Limit switches: REIFENG mechanical endstop selected (SPDT C/NO/NC, 1 A 125 VAC). Generic 3D-printer part with no formal datasheet — specs read directly from the switch body.

Components List

#	Component	Qty	Datasheet	Notes
1	ESP32 DevKit V1 (WROOM-32)	1	esp32_datasheet_en.pdf	3.3 V logic (also esp-dev-kits-en-master-esp32.pdf)
2	BTS7960 43A H-Bridge module	1	BTS7960 Motor Driver.pdf	Motor driver, 3.3–5 V logic, 6–27 V motor supply
3	DC worm gear motor, 12 V (5882-50ZY)	1	vendor spec (model identified)	Brushed, self-locking; stall ~15–19 A; confirm gear-ratio variant — needs ~200 kg·cm / 20 N·m for wind
4	P160 16 mm rotary potentiometer	1	p160.pdf	Use linear (B) taper ; 10 kΩ; 260° travel
5	DS3231 RTC module	1	DS3231.pdf	I2C 0x68, CR2032 backup
6	REIFENG mechanical endstop (SPDT microswitch on PCB, C/NO/NC, 1 A 125 VAC, 1 M cable)	2	generic 3D-printer part (no datasheet)	East & west end-stops; use C + NO contacts
7	QS-FS wind speed sensor (C2192-002)	1	C2192-002_datasheet.pdf	Powered 5–24 V sensor, NOT a reed switch — see anemometer section
8	LDR / photocell (SparkFun SEN-09088)	2	SEN-09088.pdf	Diagnostic only; 2-terminal, no polarity
9	10 kΩ resistor	2	—	LDR voltage divider pull-downs
10	10 kΩ + 20 kΩ resistor	1 ea	—	Anemometer 5 V→3.3 V signal divider (see note)
11	12 V 20 A power supply (e.g. ALITOVE 240 W)	1	—	Motor rail; 20 A covers the ~19 A stall surge
12	LM2596 buck converter (12 V→5 V, adj., 3 A)	1	—	Makes the 5 V rail; set to 5.0 V before connecting ESP32
13	12 AWG wire	—	—	Motor power path only (supply↔driver↔motor)
14	18–22 AWG wire	—	—	All logic/signal/ESP32 wiring
15	Schottky diode (1N5822 3 A, or 1N5817)	1	—	In buck 5 V output → ESP 5V pin (stripe toward ESP); lets USB stay safely pluggable for programming
—	Renogy 50 W solar panel (RNG-50D-SS)	1	50D-SS-Datasheet.pdf	The tracked panel (Voc 21.8 V, Isc 3.1 A); not wired to the controller

Common ground: ESP32 GND, BTS7960 logic GND, anemometer ground, and the 12 V supply GND must all be tied together.

Power — two rails

A single 12 V 20 A supply feeds the motor directly and an LM2596 buck converter that produces the 5 V logic rail.

```

12V 20A supply
|-- B+/B- -----> BTS7960 --> motor (M+/M-)      [12 AWG, ~19 A]
|-- buck --[Schottky]-> 5V rail ----> ESP32 5V pin
                                |-- BTS7960 VCC
                                +--- anemometer power
USB port = programming only (the diode lets you plug it in anytime)
ESP32 3.3V --> RTC, pot, LDRs, limit switches      [18-22 AWG]
ALL grounds tied together (12V supply, buck, ESP32, driver, sensors)

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From	To	Wire	Notes
12 V supply (+)	BTS7960 B+	12 AWG	Motor power rail
12 V supply (−)	BTS7960 B−	12 AWG	Common GND
12 V supply (+/−)	LM2596 IN+ / IN−	18–22 AWG	Buck converter input
Buck OUT+	Schottky diode → ESP32 5V pin, BTS7960 VCC, anemometer power	18–22 AWG	5 V logic rail; diode lets USB coexist
Buck OUT−	ESP32 GND (common)	18–22 AWG	
ESP32 3.3 V	Potentiometer pin 1	22 AWG	Pot reference voltage
ESP32 3.3 V	DS3231 VCC	22 AWG	RTC power (3.3 V — see RTC safety note)
ESP32 GND	All peripheral GNDs	22 AWG	Common GND — must be shared

- If using an adjustable buck, **set it to 5.0 V before wiring it to the ESP32.**
- Do **not** feed 12 V into the ESP32 VIN; power it from the regulated 5 V rail.

Programming: the ESP32 is powered through its **5V pin** from the buck — the USB port is for flashing code only. A **Schottky diode** in the buck’s 5 V output means you can leave the whole system powered and plug a USB cable into the computer anytime, with any cable, with no conflict and no risk to the USB port (the diode blocks any backfeed toward the computer).

BTS7960 Motor Driver → ESP32

BTS7960 Pin	ESP32 GPIO	Config symbol	Description
RPWM	GPIO 25	PIN_MOTOR_RPWM	PWM — drives motor CW (panel tilts west)
LPWM	GPIO 26	PIN_MOTOR_LPWM	PWM — drives motor CCW (panel tilts east)
R_EN	GPIO 27	PIN_MOTOR_REN	Enable right half-bridge (driven HIGH)
L_EN	GPIO 14	PIN_MOTOR_LEN	Enable left half-bridge (driven HIGH)
VCC	ESP32 5 V	—	Logic supply for the module
GND	ESP32 GND	—	Logic ground
R_IS	—	—	Current-sense output, not connected
L_IS	—	—	Current-sense output, not connected

3.3 V logic confirmed by datasheet. The BTS7960 module datasheet states “**Control Input Level: 3.3~5V**” — the ESP32’s 3.3 V GPIO drives RPWM/LPWM/R_EN/L_EN directly, no level shifter needed. Motor supply (B+/B−) range is **6–27 VDC**, VCC is +5 V.

BTS7960 → Motor & Power Terminals

Screw-terminal block labels on the IBT-2 module (verified against datasheet/vendor docs).

Terminal	Connection	Notes
M+	Motor wire A	Swap A/B if the panel drives the wrong direction
M−	Motor wire B	
B+	12 V supply (+)	Motor power rail, 6–27 V
B−	12 V supply (−)	Common ground

Power sizing: the 5882-50ZY draws ~**15–19 A at stall** (startup inrush or a jam), though normal slow tracking moves draw far less. The BTS7960 (43 A) handles this easily, but the **12 V supply/battery needs ≥15–20 A surge capability** — a 10 A supply may brown out on stall. Fuse ~10–15 A on B+; 18 AWG wire is near its limit at stall current.

Potentiometer (Shaft Angle Feedback)

Pot pin	Connection	Notes
Pin 1 (end)	ESP32 3.3 V	Reference high
Pin 2 (wiper)	ESP32 GPIO 34	PIN_POT — ADC1_CH6
Pin 3 (end)	ESP32 GND	Reference low

Use ADC1 pins only (GPIO 32–39). ADC2 is disabled while WiFi is active. Run `runCalibration()` on first assembly to find POT_ADC_MIN / POT_ADC_MAX.

Limit Switches — REIFENG mechanical endstop (SPDT, C/NO/NC)

One switch at each end of travel. The board has an SPDT microswitch (terminals **C** / **NO** / **NC**, rated 1 A 125 VAC) on a small PCB with mounting holes and a 3-wire cable (red/black/white). Firmware uses INPUT_PULLDOWN + LIMIT_ACTIVE HIGH, so the switch must connect the GPIO to **3.3 V** when pressed — use the **C** and **NO** contacts.

Switch	ESP32 GPIO	Config symbol	Triggers when...
CW (west) limit	GPIO 32	PIN_LIMIT_CW	Panel reaches full west travel
CCW (east) limit	GPIO 33	PIN_LIMIT_CCW	Panel reaches full east travel

Switch contact	Connect to
C (common)	ESP32 3.3 V
NO (normally open)	ESP32 GPIO 32 (CW) or 33 (CCW)
NC (normally closed)	not connected

Identify the wires (red/black/white order isn't guaranteed) with a multimeter in continuity mode, or with the bring-up tool:

1. Lever **released** → the two wires that beep are **C and NC**.
2. Lever **pressed** → the two wires that beep are **C and NO**.
3. The wire common to both is **C** (→ 3.3 V); the other from step 2 is **NO** (→ GPIO).

Not weather-sealed — mount inside the electronics enclosure for a permanent outdoor install. (To wire C+NC instead, change `LIMIT_ACTIVE` to `LOW` and the `pinMode` to `INPUT_PULLUP`.)

Anemometer (Wind Speed) — QS-FS, part C2192-002

! This is NOT a passive reed switch. Per [datasheets/C2192-002_datasheet.pdf](#), the QS-FS is a **powered electromagnetic-induction sensor** that needs its own supply and outputs an actively-driven signal. This changes the wiring substantially from the earlier reed-switch assumption.

Specs: Supply 5–24 VDC · output options (variant-dependent): **Pulse (5 V)**, 4–20 mA, 0–5 V/1–5 V, or RS485 · range 0–60 m/s · starting threshold ≤ 0.8 m/s · IP55.

3-wire connection (Pulse / current / voltage variants):

Function	Wire color (2 schemes)	Connect to
Power	red or brown	ESP32 5 V (VIN) — sensor needs 5–24 V, use 5 V
Ground	blue or black	ESP32 GND (common)
Signal	yellow or blue	GPIO 35 (<code>PIN_ANEMOMETER</code>) via level shift — see below

Two things to resolve before wiring this:

1. **Confirm which output variant you bought.** The current firmware counts digital pulses (RISING-edge interrupt), so it only works with the **Pulse (5 V)** variant. If you have the 4–20 mA, 0–5 V, or RS485 version, the firmware and wiring must change (ADC + sense resistor, or a MAX485/UART transceiver). The “-002” suffix likely encodes the variant — verify with the supplier.
2. **5 V signal must be dropped to 3.3 V.** The pulse output is **5 V**, but GPIO 35's absolute max is ~ 3.6 V — driving it at 5 V can damage the pin. Use a **voltage divider** on the signal line: signal → **10 k Ω** → GPIO 35, and GPIO 35 → **20 k Ω** → GND (gives $5\text{ V} \times 20/30 \approx 3.3\text{ V}$). A logic-level shifter also works.
3. **Recalibrate ANEM_MPH_PER_HZ.** The default 1.49 was for a SparkFun reed-switch anemometer and is wrong for the QS-FS. Get the pulse-frequency-to-wind-speed relation from the QS-FS manual/supplier and update `config.h`.

LDR Sensors (Diagnostic)

Voltage divider: $3.3\text{ V} \rightarrow \text{LDR} \rightarrow \text{ADC pin} \rightarrow 10\text{ k}\Omega \rightarrow \text{GND}$. Used only to cross-check the astronomical algorithm — not required for tracking.

LDR	ESP32 GPIO	Config symbol	Position
East LDR	GPIO 36	PIN_LDR_EAST	Faces east (morning sun)
West LDR	GPIO 39	PIN_LDR_WEST	Faces west (afternoon sun)

Node	Connection
LDR top	ESP32 3.3 V
LDR bottom / ADC junction	ESP32 GPIO 36 or 39
10 k Ω resistor (top)	ADC junction
10 k Ω resistor (bottom)	ESP32 GND

DS3231 RTC Module (I2C)

DS3231 pin	ESP32 GPIO	Config symbol	Notes
VCC	3.3 V	—	Power from 3.3 V — see safety note below
GND	GND	—	
SDA	GPIO 21	RTC_SDA	I2C data, address 0x68
SCL	GPIO 22	RTC_SCL	I2C clock
SQW / INT	—	—	Not connected
32K	—	—	Not connected

Power from 3.3 V, not 5 V (safety). The common ZS-042 DS3231 board has an onboard battery-charging circuit (diode + $200\ \Omega$) intended for a rechargeable LIR2032. If you power VCC at 5 V while using a **non-rechargeable CR2032**, it pushes $\sim 5\text{ mA}$ into a cell that cannot take a charge — risk of overheating/leakage. Powering from 3.3 V keeps the diode reverse-biased and prevents charging. (If you must run 5 V, fit a LIR2032 or remove the charging resistor.)

- The DS3231 chip's I2C address is **0x68**; the onboard EEPROM is 0x57 (or 0x50–0x57). The bring-up I2C scan should report 0x68.
- Install a CR2032 coin cell so the RTC keeps time through power loss.
- The module includes onboard I2C pull-up resistors (typically $4.7\text{ k}\Omega$) — do not add external pull-ups.

ESP32 GPIO Summary

GPIO	Function	Direction	Notes
14	Motor L_EN	Output	BTS7960 left enable
21	I2C SDA	Bidirectional	DS3231 RTC
22	I2C SCL	Output	DS3231 RTC
25	Motor RPWM	Output	PWM CW
26	Motor LPWM	Output	PWM CCW
27	Motor R_EN	Output	BTS7960 right enable
32	Limit CW	Input (pull-down)	West end-stop
33	Limit CCW	Input (pull-down)	East end-stop
34	Potentiometer	ADC1 input	Shaft angle feedback
35	Anemometer	Input-only	QS-FS pulse — 5 V signal, must divide to 3.3 V (10 k/20 k)
36	LDR East	ADC1 input	Diagnostic
39	LDR West	ADC1 input	Diagnostic